

Scattering part 1

Let $\psi(r, \theta, \phi)$ be a wave function of the form

$$\psi(r, \theta, \phi) = \frac{e^{ikr}}{r} f(\theta, \phi)$$

where

$$k = \frac{\sqrt{2mE}}{\hbar}$$

The density is

$$|\psi|^2 = \psi^* \psi = \frac{|f(\theta, \phi)|^2}{r^2}$$

Let $d\sigma$ be the differential cross section

$$d\sigma = |\psi|^2 dA = |\psi|^2 r^2 d\Omega = |f(\theta, \phi)|^2 d\Omega$$

Hence

$$\frac{d\sigma}{d\Omega} = |f(\theta, \phi)|^2$$

For Coulomb potentials we have

$$f(\theta, \phi) = \frac{Z\alpha\hbar c}{2E(1 - \cos\theta)}$$

where Z is the dimensionless integer

$$Z = \frac{q_1 q_2}{e^2}$$

Hence the Rutherford cross section

$$\frac{d\sigma}{d\Omega} = \left[\frac{Z\alpha\hbar c}{2E(1 - \cos\theta)} \right]^2$$

Noting that

$$(1 - \cos\theta)^2 = 4 \sin^4(\theta/2)$$

we can also write

$$\frac{d\sigma}{d\Omega} = \left(\frac{Z\alpha\hbar c}{2E} \right)^2 \frac{1}{4 \sin^4(\theta/2)}$$

Verify dimensions of exponential.

$$\frac{\sqrt{2mE}}{\hbar} r = \frac{\sqrt{[\text{kg}] [\text{J}]}}{[\text{J s}]} [\text{m}] = [1] \quad (1)$$

Verify dimensions of cross section.

$$\left(\frac{Z\alpha\hbar c}{2E} \right)^2 = \left(\frac{[\text{J s}] [\text{m s}^{-1}]}{[\text{J}]} \right)^2 = [\text{m}^2] \quad (2)$$

Eigenmath script